# OLUWATOONI JOEL, ADEBAYO

joeltooni@cmu.edu | linkedin.com/in/joeltooni | (+250) 794007573

## RESEARCH INTERESTS

Object Recognition and Scene Understanding | Sensor Fusion for Autonomous Robots in Dynamic Environments | Reinforcement Learning for Robot Control | Artificial Internet of Things (AIoT) | Simultaneous Localization and Mapping (SLAM) | Robot Learning from Demonstration (LfD) | Vision Language Models (VLMs)

## EDUCATION

**Master of Science in Electrical & Computer Engineering** **June 2024 – Present**
Carnegie Mellon University, Africa **CGPA:** 3.71/4.0 (92.7%)
**Relevant Coursework:** Robotics Principles and Practice, Applied Computer Vision, Data Inference and Applied Machine Learning, Deep Learning, Humanoid Robotics & Cognition, Engineering Research Methods, Applied Stochastic Processes

**Bachelor of Science in Systems Engineering** **December 2015 - December 2021**
University of Lagos, Akoka, Nigeria **CGPA:** 4.51/5.0 (90.2%)
**Relevant Coursework:** Image Processing, Algorithms & Data Structures, Mechanics of Robotics Systems I & II, Artificial Intelligence, Math. Modeling for AI Systems, Stochastic Models, Control Theory I & II

## SKILLS

- **Microcontrollers:** Arduino, Raspberry Pi, ESP-32, ESP-32 CAM
- **Programming:** C, C++, Python, Java
- **Robotics:** ROS, Fritzing, Fusion 360, OpenCV, Linux/Unix, Ultimaker Cura
- **Data Analysis/ML:** Sci-kit-learn, NumPy, Pandas, Matplotlib, PyTorch, Matplotlib, Seaborn
- **Others:** Anaconda, Git, GitHub, Docker, Festo Fluid Sim, Tora, VSCode, Jupyter Notebook, Zotero, Latex

## EXPERIENCES

**Research Assistant** **July 2023 - March 2024**
*Computer Engineering Department, Nigerian Army College of Environmental and Sci,* Makurdi, Nigeria.

- Created a user interface for real-time monitoring of eight sensors in a Smart Armory, enhancing operational efficiency.
- Worked on the pipeline that achieved 85% accuracy in image recognition and classification, integrating sensor data with AI algorithms to address a problem faced by the Nigerian military.

**Course lecturer/IT Support** **July 2022 - June 2023**
*Computer Engineering Department, Nigerian Army College of Environmental and Sci,* Makurdi, Nigeria.

- Prepared course materials, exercises, exams and quizzes and taught 12 classes for Computer Application Packages (CTE 113) & Internet and Web Technologies (CTE 114) to a class size of 11, strictly adhering to provided curriculum.
- Created laboratory manuals and led eight (8) experiments for two courses.
- Adopted active learning methodologies and introduced "office hours" - 27% of students keyed into it.
- Provided IT support to over 300 students in 3 months, drastically reducing long queues at the ICT unit.

**Research Assistant & Lab Assistant** **March 2020 - April 2022**
*Artificial Intelligence and Robotics Laboratory (AiRoL), University of Lagos,* Lagos, Nigeria.

- Managed laboratory operations and maintained an up-to-date inventory of kits and equipment.
- Devised recruitment programs for coding team competitors and mentors.
- Facilitated selecting and training team members to compete in various competitions: FLL 2019/2020 Session, FLL & FTC 2020/2021 (4 local & 2 international awards), Volvo hauler challenge (3rd place).
- Planned and executed 3 undergraduate coding and robotics workshop/internship programs.

## PROJECTS

**Robotic Modeling of the Nigerian Cooking Tradition – Jollof Rice Cooking Robot** **January 2025 - Present**

- Working in a team of 4 under the supervision of Prof Conrad Tucker and we a developing a custom robotic arm that perceives ingredients, follows expert cooking steps, and executes tasks using multimodal sensor fusion and vision-based control
- Integrating computer vision and AI to enable real-time decision-making for precise cooking automation.

**6DoF Object Pose Estimation using Multi-View and Multimodal Industrial Plenoptic Datasets** **January 2025 - Present**

- Working in a group of 3 and we a developing a 6-degree-of-freedom (6DoF) object pose estimation pipeline to enhance robotic perception and manipulation in industrial settings.
- Currently developing the deep learning pipeline to accurately detect object poses while ensuring generalizability across multiple test parts.

**Path Planning and Mobile Robot Navigation with ROS** **August 2024 - November 2024**

- Designed and implemented a path planning ROS program for the iRobot (Roomba) to navigate a 2D grid-based map with obstacles, utilizing breadth-first search and Dijkstra's algorithms to compute shortest path while respecting robot configuration constraints.
- Displayed planned paths in binary output images superimposed on workspace and configuration space maps and controlled the iRobot via a ROS program to follow computed path and waypoints after planning.

**Robot Arm Control for Brick Manipulation** **August 2024 - November 2024**

- Developed a control program for the LynxMotion AL5D robot arm to pick up LEGO bricks by specified start and goal Cartesian coordinates from an input file and homogenous transformations.
- Automated detection of bricks employing OpenCV and a USB camera and incorporating inverse perspective transformation to accurately determine positions and orientations.

## PUBLICATIONS

- Olurotimi A. Adeleye, and **Oluwatooni J. Adebayo** (2024) Design and Implementation of a Smart Waste Management System using AIoT in Mitigating Poor Air Quality Intake and contraction of diseases in a closed community. Journal of Engineering Research - under publishing review.
- Chika Yinka-Banjo, Sunday Adebisi, and **Joel Adebayo** (2021) Rescuing African Youth Future from Unemployment through Early Strategic Training of Adolescents in Coding, AI, and Robotics for Technopreneurship Opportunities. Journal of African Employment, Entrepreneurship & Skills Development (JAEESD) Vol.1 (3), 73-85.

## PRESENTATIONS

- **Speaker: A-Z of Landing that Dream Job, I**mpart Academy Workshop, Makurdi, Nigeria. **October 2023**
- **Resource Person,** AI Summit, Nigerian Computer Society (NCS). **April 2023**
- **Image Restoration and Reconstruction with MATLAB,** SSG512 Class presentation. **September 2021**

## LEADERSHIP AND VOLUNTEERING EXPERIENCES

- **Chair,** *IEEE Student Branch*, CMU, Africa. **January 2025 - Present**
- **Community Engagement Officer,** *Research Club,* CMU, Africa. **November 2024 - Present**
- **Volunteer Instructor,** *Experience A.I,* Kigali, Rwanda. **November 2024**
- **Instructor,** *Summer Coding Outreach, NAF Base,* Makurdi, Nigeria**.** **August 2023 - September 2023**
- **Robot Coach/Mentor,** *First Lego League, Team AiRoL - AiRoL UNILAG***,** Nigeria. **2019 – 2021**

## MEMBERSHIPS AND AFFILIATIONS

- **Member**, IEEE Students **February 2025 - Present**
- **Member**, Black in Artificial Intelligence (BAi). **August 2023 - Present**
- **Member and Teaching Assistant**, Black in Robotics (BiR). **July 2023 – Present**

## CERTIFICATIONS

- **Machine Learning Fundamentals in Python (Skill Track),** Datacamp. **January 2025**
- **Python 3 Programming Specialization,** University of Michigan. **May 2023**

- **Introduction to Embedded Machine Learning,** Edge Impulse. **November 2022**
- **An Introduction to Programming the Internet of Things,** University of California, Irvine. **January 2022**
- **An Introduction to Evidence-Based Undergraduate STEM Teaching,** CIRTL. **November 2019**

## AWARDS, HONORS, AND RECOGNITIONS

- **Beneficiary**, IDEA Travel Support Grant by IEEE Women in Engineering **February 2025**
- **1ST PLACE,** e-Waste Challenge, EPRON. **October 2024**
- **3RD PLACE**, Tartan Innovate Challenge. **September 2024**
- **Mastercard Foundation Scholar,** Carnegie Mellon University, Africa. **April 2024**
- **2ND RUNNER UP,** Business Startup Grant Competition for Engineering Students. **2021**
- **4TH PLACE**, The Coach/Mentor Award First® Lego® League Challenge, Open International, Greece. **2021**
- **Beneficiary**, Wilson Badejo Foundation Scholarship. **2016 & 2017**